

Project 1

Trapped charged particle Problem

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NUMERICAL MODELING OF TRAPPED CHARGED PARTICLES IN THE MAGNETOSPHERE

PROJECT OVERVIEW AND LEARNING OBJECTIVES

This project outlines the requirements for developing a high-fidelity numerical simulation of charged particle dynamics within the Earth's magnetic dipole field. As project in computational astrophysics, the objective is to move beyond simple integration and achieve a rigorous synthesis of numerical analysis and magnetospheric physics. You are tasked with modelling the "bounce" and "drift" motions of particles trapped in the Van Allen radiation belts.

The core learning objectives are:

- Numerical Differentiation Theory: Mastery of Lagrange interpolation-based differentiation and the rigorous calculation of truncation errors.
- Iterative Nonlinear Solvers: Implementation of the Bisection Method to resolve critical transition points (mirror points) in particle trajectories.
- Physical Stability & Validation: Utilizing first adiabatic invariants (μ) and conservation of energy to validate the numerical integrity of the simulation over extended temporal scales (10^5 iterations).
- Algorithmic Generalization: Developing modular code that maps cofactor determinants to finite difference coefficients.

PHASE I: MATHEMATICAL FRAMING OF PARTICLE DYNAMICS

The student must derive the equations of motion for a charged particle subjected to the Lorentz force, $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. For this simulation, assume a static magnetic dipole field and zero electric field. The dynamics must be discretized using finite difference approximations for the velocity and acceleration components.

NUMERICAL DIFFERENTIATION SPECIFICATION

Following the general differentiation frameworks established by Singh and Thorpe, you must implement finite difference approximations derived from Lagrange interpolation.

- Order of Approximation: Implement the general differentiation formula for $n = 2$ and $n = 3$ nodal points.

- Second Derivative Implementation: To represent the acceleration components $\mathbf{a} = \frac{d^2\mathbf{r}}{dt^2}$ in the Lorentz force equation, implement the second derivative in its central difference form:

$$f''(x) = \frac{f_{i-1} - 2f_i + f_{i+1}}{h^2} \quad (0.1)$$

where h is the simulation time step and function values are indexed according to their temporal grid position.

PHASE II: ALGORITHMIC GENERALIZATION AND LAGRANGE INTERPOLATION

The simulation requires a generalized algorithm for function approximation. This is critical for determining field values at arbitrary particle coordinates between established grid points.

LAGRANGE IMPLEMENTATION

Implement the general Lagrange interpolation formula:

$$f(x) = \sum_{j=0}^n l_j(x) f_j \quad (0.2)$$

where the basis polynomials are defined by the product term $\pi(x) = (x - x_0)(x - x_1) \dots (x - x_n)$:

$$l_j(x) = \frac{\pi(x)}{(x - x_j)\pi'(x_j)} \quad (0.3)$$

RIGOROUS ERROR ANALYSIS

To maintain graduate-level precision, you must distinguish between interpolation and differentiation errors. While the interpolation error is $E_n(x)$, you are required to calculate the differentiation error E'_n specifically for your force calculations. Citing Singh and Thorpe, the error of approximation for the derivative is:

$$E'_n = \frac{h^n}{(n+1)!} \cdot \frac{f^{n+1}(\xi)}{a_{0n}} \left\{ \sum_{k=0}^n (k+1) a_{0k} s^k \right\} \quad (0.4)$$

You must demonstrate how your choice of nodal points n affects the stability of the particle's trajectory over long-term integration.

PHASE III: UNIT HANDLING AND PHYSICAL PARAMETERS

The simulation shall be conducted in a normalized unit system to ensure computational efficiency and prevent floating-point instability. The following mandatory constants, derived from the project context, must be integrated into the dipole model:

- Initial L -shell (L): 8 (Initial radial distance in Earth radii).
- Magnetic Field Constant (B_0): 2.667 (Normalized field strength factor).
- Charge-to-Mass Ratio (q/m): 4.
- Initial Kinetic Energy Scalar (E_{k0}): 7.

PHASE IV: NUMERICAL INTEGRATION AND METHODOLOGICAL COMPARISON

Temporal evolution must be calculated using a finite difference-based integration routine (e.g., 4th-order Runge-Kutta or a Velocity Verlet scheme).

COMPARATIVE ANALYSIS OF DIFFERENTIATION SCHEMES

You are required to submit a comparative analysis of local truncation errors (LTE) for first-derivative approximations. These values must be derived by setting the parameter $s = 0, 1, 2$ in the general error formula for $n = 2$. Students must provide a mathematical justification, based on these LTE values, for the selection of the central

Difference Scheme	s	value Order (n)	Local Truncation Error (LTE)
Forward Difference	0	2	$\frac{h^2 f'''(\xi)}{2}$
Backward Difference	2	2	$\frac{h^2 f'''(\xi)}{2}$
Central Difference	1	2	$-\frac{h^2 f'''(\xi)}{6}$

difference scheme to minimize numerical drift in periodic orbits.

PHASE V: PHYSICAL VALIDATION AND ADIABATIC INVARIANTS

The physical validity of the simulation must be rigorously checked via the conservation of the first adiabatic invariant (magnetic moment, μ).

THE BISECTION METHOD FOR MIRROR POINTS

Implement the Bisection Method to solve for the particle's "mirror points". These are the latitudes where the parallel velocity $v_{\parallel}$ reaches zero. You must solve the nonlinear equation $f(\lambda) = v_{\parallel}(\lambda)$ for the root λ_m (the magnetic latitude of the mirror point).

STABILITY AND INTERPOLATION

The simulation must run for 10^5 iterations with a maximum allowed energy drift of 10^{-6} . To monitor stability, calculate:

1. Total Energy (E): $E = E_{\perp} + E_{\parallel}$.
2. Magnetic Moment (μ): $\mu = \frac{E_{\perp}}{B}$.
3. Requirement: Since the particle will rarely be on a grid point, you must use the Lagrange interpolation formula from Phase II to determine the local magnetic field strength B at the particle's exact position.

VISUALIZATION AND DELIVERABLES

The final submission must include detailed descriptions (not the files themselves) of the following plots:

- Uniform magnetic field. Follow charged particles under a uniform magnetic field $B = B\hat{z}$ where $B = 1\text{T}$. Verify that the particles follow helices with cyclotron radius and frequency. Experiment with particles with different mass and charge values.
- Gradient drift. Consider a magnetic field given as $B = (A\hat{x} + B_0\hat{z})_z$. The field has a gradient in the x -direction, but no curvature. Set $A = 1\text{Tm}^{-1}$ and $B_0 = 1\text{T}$. Follow the trajectory of a particle with mass $m = 1\text{kg}$ and $q = 1\text{C}$ initialized with velocity $v = 1\text{ms}^{-1}\hat{y}$ at the origin. Note that the sideways drift arises from the fact that the cyclotron radius is smaller at stronger fields.
- Equatorial particles. Consider a particle in a dipolar magnetic field, located at the equatorial plane ($z = 0$) with zero parallel speed. As the field strength is minimum at the equator with respect to the field line, there is no parallel acceleration and the particle stays on the equatorial plane at all times. Using the dipole model, follow an equatorial particle and verify that the center of the motion stays on a contour of constant B , as implied by the conservation of the first adiabatic invariant.

- Explore the drift motion. Run the programs in the supplement to trace protons and electrons using the dipole model. Initialize particles with different energies, starting positions and pitch angles. Verify that electrons and protons drift in opposite directions, and electrons have a much smaller cyclotron radius than protons with the same kinetic energy. Estimate the periods of bounce and drift motions
- Accuracy of the guiding-center approximation. Simulate the full particle and guiding center trajectories with the same initial conditions and plot them together. Shift the initial position of the particle properly so that the guiding center runs through the middle of the helix. Repeat with protons with 1keV, 10keV, 100keV and 1MeV kinetic energies. At higher energies, the guiding-center trajectory lags behind the full particle because the omitted high-order terms become more significant as the cyclotron radius increases.
- Different numerical methods. Solve the full particle and guiding center equations using different numerical schemes, such as Verlet, EulerCromer, Runge-Kutta and Bulirsch-Stoer. Verify the accuracy of the solution by checking the conservation of kinetic energy and adiabatic invariants.

GENERALIZED CODE IMPLEMENTATION AND SUBMISSION REQUIREMENTS

The project submission must be modular and documented to a professional standard.

1. Documented Source Code: Original implementation of the RK4 or Symplectic solver.
2. Technical Report: A comprehensive synthesis comparing analytic and numerical results, including the formal comparison
3. All documentation must adhere to professional academic standards. Maintain technical rigour throughout and ensure all derivations are logically sequenced.